Building a RAG-based Chat Assistant

Objective:

Participants will extend their knowledge by implementing a **Retrieval-Augmented Generation (RAG)** system. They will build a **chat assistant** for a product company that leverages an embedding model and a vector database to enhance LLM responses with relevant knowledge from a given dataset.

Assignment Overview:

- Participants will develop a chatbot that retrieves relevant answers from a provided **FAQ PDF** before generating responses.
- The chatbot should **embed and index the provided FAQ document** using an example embedding model and vector database.
- The assistant should **first retrieve relevant context from the document** before generating a response with the LLM.
- The UI should allow users to ask product-related questions and receive AI-driven responses enhanced by retrieved knowledge.

Key Features to Include:

1. **PDF Data Processing:** Extract and embed the FAQ document for retrieval.
2. **Vector Search:** Implement a vector database to store and efficiently retrieve embeddings.
3. **RAG Workflow:** The chatbot should retrieve relevant passages before generating responses.
4. **User Query Handling:** Users should be able to ask freeform questions about the product.
5. **Accuracy & Relevance:** The chatbot should prioritize accuracy by providing document-backed responses.
6. Integrate Langsmith for observability

Expected Outcomes:

By completing this assignment, participants will:

- Learn **Retrieval-Augmented Generation (RAG)** techniques for enhancing LLM responses.
- Implement **embedding models** and use a **vector database** to store and retrieve knowledge.
- Build an AI chatbot that delivers **accurate, context-aware** responses based on retrieved information.

- Understand the challenges of **document processing, indexing, and retrieval-based AI systems**.

Product FAQ (Pick any one)

Product Edutrack FAQ

Product TuneHive FAQ

Product QuickSupport FAQ